UNIT 1 MINI PROJECT

DIGITAL LIBRARY MANAGEMENT PLATFORM

Design Thinking and Product Planning

Student Details	Information
Name	Gaurav Borle
PRN	2125UDSM1071
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Project Overview

The Digital Library Management Platform is a web-based academic library solution designed to replace slow, fragmented library processes with a centralized digital system. The project follows a product-engineering approach: understand the domain and users, define requirements, design the product, establish an MVP, and prepare the system for Agile implementation.

Objective

To design a modern library platform that helps students and faculty discover resources quickly while enabling librarians and administrators to manage catalog records, circulation, fines, and operational reports from one reliable system.

Project Scope

- Book catalog and inventory management.
- Student/faculty membership and role-based access.
- Book search and availability tracking.
- Issue, return and renewal workflow.
- Automated overdue fine calculation.
- Due-date notifications and basic administrative reporting.

Engineering Approach

The project adopts the lifecycle described in the training reference: requirement and domain analysis, Design Thinking and UX, architecture and database design, implementation, testing/CI-CD, and production monitoring. Coding is therefore treated as one part of a complete product-engineering lifecycle.

1. Business Requirements Document (BRD)

1.1 Purpose

This document defines the business needs for a centralized Library Management Platform for a higher-education institution. It aims to reduce manual searching, paper-based circulation records, overdue tracking effort, and reporting delays.

1.2 Problem Statement

Students and faculty face delays in locating, reserving, borrowing, and returning resources because library records are often maintained through manual or disconnected processes. Librarians also lack centralized real-time visibility into inventory, circulation, and overdue items.

1.3 Business Objectives

- Centralize catalog, membership, and circulation information.
- Reduce the time required to locate and issue books.
- Improve inventory accuracy and traceability.
- Automate overdue fine calculation and reminders.
- Provide role-based access and useful operational reports.

1.4 Scope

In Scope	Out of Scope for MVP
Book catalog and inventory	AI-based recommendations
Member profiles and RBAC	Online payment gateway
Search and availability	Full external ERP integration
Issue / return / renew	Advanced predictive analytics
Fine calculation and basic reports	Native mobile application

1.5 Business Requirements

ID	Requirement	Business Justification
BR-01	Maintain a single digital record for each book and member.	Reduces duplicate and inconsistent records.
BR-02	Provide fast multi-field book search.	Reduces physical searching time.
BR-03	Digitize issue and return transactions.	Reduces paper-ledger errors.
BR-04	Calculate overdue fines automatically.	Improves consistency and collection.
BR-05	Provide operational reports.	Supports library planning and auditability.
BR-06	Enforce role-based access.	Protects sensitive administrative data.

2. Product Vision Document

2.1 Vision Statement

To provide a reliable, user-friendly digital library experience that makes academic resources easy to discover and borrow while giving library teams accurate, centralized control over catalog and circulation operations.

2.2 Mission Statement

To simplify academic library operations through secure, maintainable software that connects catalog management, member services, circulation, fines, and reporting in one platform.

2.3 Product Positioning

Element	Description
For	Students, faculty, librarians and academic administrators
Who need	A fast and reliable way to discover and manage library resources
The product	Digital Library Management Platform
Is	A web-based library management system
That	Centralizes catalog, circulation, fines and reporting
Unlike	Paper registers and disconnected manual processes
Our product	Provides searchable, role-based and auditable library operations

2.4 Success Criteria

- Users can locate a book quickly through search and availability status.
- Librarians can complete issue/return transactions without paper ledgers.
- Overdue items and fines are visible without manual calculation.
- Administrators can obtain basic circulation and inventory reports.
- The MVP is usable, testable, and ready for subsequent Agile development.

3. Stakeholder Analysis Report

The project recognizes business, engineering and operations stakeholders so that usability, quality, security and maintainability are considered together.

Stakeholder	Role / Interest	Influence	Expectations
Students	Search, reserve and borrow books	Medium	Simple search, availability and due-date information
Faculty	Access academic references efficiently	Medium	Fast search and reliable borrowing
Librarians	Manage catalog and circulation	High	Quick issue/return and accurate inventory
Library Administrator	Monitor utilization and reports	High	Centralized visibility and auditability
IT / DevOps	Deploy and maintain platform	Medium	Secure, modular and maintainable system
College Management	Reviews service efficiency	High	Measurable improvement in library operations

3.1 Stakeholder Engagement

- Students and faculty: gather usability needs through interviews or short surveys.
- Librarians: observe cataloging, search, checkout and return workflows.
- Administrators: validate reporting and policy requirements.
- Engineering/IT: validate security, performance, deployment and maintainability constraints.

4. User Persona Report

4.1 Persona: The Student

Attribute	Details
Name / Role	Rohan Mehta, Undergraduate Student
Goals	Find references quickly, borrow books, avoid overdue fines.
Challenges	Long queues, difficulty knowing availability, unexpected overdue fines.
Expectations	Mobile-friendly search, availability status and due-date reminders.

4.2 Persona: The Librarian

Attribute	Details
Name / Role	Ms. Kavita Sharma, Librarian
Goals	Process checkouts quickly and keep inventory accurate.
Challenges	Manual entries, stock discrepancies and lost records.
Expectations	Fast issue/return interface, automated fine tracking and catalog controls.

4.3 Persona: The Library Administrator

Attribute	Details
Name / Role	Mr. Amit Kulkarni, Library Administrator
Goals	Monitor usage and generate departmental reports.
Challenges	No real-time usage dashboard and time-consuming report preparation.
Expectations	Executive dashboard, circulation metrics and export-ready reports.

5. Functional Requirements Specification (FRS)

The following requirements define the core behavior of the MVP.

ID	Functional Requirement	Priority
FR-01	Secure role-based login for Admin, Librarian, Student and Faculty.	High
FR-02	Authorized staff can create, update, view and deactivate book records.	High
FR-03	Search by title, author, category, ISBN and availability.	High
FR-04	Show current stock/availability for each book.	High
FR-05	Record issue transactions and decrement available inventory.	High
FR-06	Record returns and increment available inventory.	High
FR-07	Calculate overdue fines using configured daily rate and overdue days.	High
FR-08	Allow authorized users to renew eligible loans.	Medium
FR-09	Generate due-date and overdue notifications.	Medium
FR-10	Generate circulation, overdue and inventory reports.	Medium
FR-11	Maintain transaction history for audit purposes.	Medium

6. Non-Functional Requirements Specification (NFRS)

Category	Requirement
Security	Role-based access, secure authentication, hashed passwords and HTTPS in deployment.
Usability	Core search and issue/return tasks should be understandable to first-time users.
Performance	Book search should return results within a few seconds under normal usage.
Reliability	Issue/return transactions must not lose or duplicate inventory records.
Scalability	Support growth in books, members and transactions without redesigning core modules.
Availability	Target high availability during library operating hours.
Maintainability	Modular separation between UI, API, business logic and database.
Auditability	Critical changes and circulation actions should be traceable.
Data Integrity	Database constraints should prevent invalid users, books and circulation states.

Technical Quality Targets

- Target sub-500 ms response for common search APIs under normal test load.
- Use relational database integrity constraints for core records.
- Containerization and CI/CD can be introduced as the project evolves.
- Monitoring can be added in later deployment phases.

7. Design Thinking Report

The project follows the five Design Thinking stages: Empathize, Define, Ideate, Prototype and Test.

Stage	Application to the Library Platform
Empathize	Observe students, faculty and librarians to understand search, borrowing, return and reporting pain points.
Define	Frame the central problem as inefficient manual tracking causing delays, errors and poor visibility.
Ideate	Explore searchable catalog, QR membership, reminders, self-service options, live dashboards and automated fines.
Prototype	Create low-fidelity screens for login, search, librarian checkout and admin analytics.
Test	Review prototype flows with representative users and improve navigation, clarity and task completion speed.

Core Problem Statement

Library members experience delays in locating, reserving, borrowing and returning resources because of inefficient manual tracking, while library staff lack centralized real-time visibility into inventory and circulation.

8. IBM Design Thinking Report

8.1 Focus on User Outcomes

Technical goals are translated into measurable user outcomes. Instead of simply building a search query, the desired outcome is to help a student locate and identify an available academic book with minimal steps.

8.2 Restless Reinvention

The MVP is treated as a starting point. Usage data, search behavior, transaction delays and user feedback can be used to continuously improve the interface and backend workflows.

8.3 Diverse Empowered Teams

The solution benefits from collaboration among product owners, UX designers, developers, QA engineers, database administrators, librarians and DevOps personnel.

Feature Goal	User Outcome
Build a catalog search service	Enable students to find an available book quickly
Create checkout screen	Enable librarians to issue a book with minimal data entry
Create reporting dashboard	Enable administrators to see circulation and overdue trends

9. IBM Loop Documentation — Observe → Reflect → Make

Iteration	Observe	Reflect	Make
1 — Search	Students spend time asking staff whether books are available.	A searchable catalog can reduce dependency on physical searching.	Define search and availability requirements.
2 — Circulation	Paper/manual checkout creates queues and data-entry errors.	Digital issue/return can update inventory immediately.	Define circulation workflow and transaction records.
3 — Overdue Tracking	Staff manually identify overdue books and calculate fines.	Automated calculation and alerts can improve compliance.	Define fine engine and notification requirements.
4 — Reporting	Administrators spend time compiling usage information.	Centralized transactions can produce reports automatically.	Define inventory, circulation and overdue reports.

10. Minimum Viable Product (MVP) Definition

10.1 MVP Goal

Deliver the smallest usable library platform that replaces the most painful manual processes: book discovery, circulation tracking, overdue calculation and basic reporting.

10.2 MVP Scope

- Role-based login for Admin, Librarian, Student and Faculty.
- Book catalog CRUD for authorized staff.
- Book search and availability status.
- Issue and return transaction management.
- Basic automated fine calculation.
- Basic inventory, circulation and overdue reports.

10.3 Explicitly Deferred

- AI book recommendations.
- Online payment gateway.
- Full institutional ERP integration.
- Advanced predictive analytics.
- Native mobile application.

10.4 MVP Success Criteria

- A student can search and identify available resources without visiting the circulation desk.
- A librarian can issue and return a book quickly.
- An overdue transaction produces a consistent fine amount.
- An administrator can view basic circulation and inventory reports.

11. Product Roadmap

Phase	Focus	Key Capabilities
Phase 1 — Foundation (MVP)	Core library records	Login, catalog, search, issue/return, fines, basic reports
Phase 2 — Workflow Enhancement	Better circulation	QR/barcode support, renewals, automated reminders, reservations
Phase 3 — Analytics	Institutional insight	Usage dashboards, loss analysis, replenishment reports, exports
Phase 4 — Integration	Ecosystem connectivity	Email/SMS, ERP integration, cloud deployment and advanced monitoring

11.1 Agile Alignment

The project can be developed incrementally through short sprints. Each sprint should produce a demonstrable increment, collect feedback, and update the backlog based on user outcomes.

12. Initial System Architecture

The proposed platform uses a layered architecture so presentation, business logic and data concerns remain independently maintainable.

Layer	Responsibilities
Presentation Layer	Responsive student/faculty portal, librarian dashboard and admin console.
API / Application Layer	Authentication, authorization, request routing and API endpoints.
Business Logic Layer	Book, user, circulation, fine, notification and reporting services.
Data Access Layer	ORM/repositories, validation, database transactions and integrity rules.
Database Layer	Relational storage for users, books, categories, loans, fines and audit records.
External Services	Email/SMS, cloud storage and other integrations can be connected later.

High-Level Flow

User Interface → REST API → Authentication & Business Services → Data Access Layer → Relational Database

13. Project Charter

Field	Details
Project Name	Digital Library Management Platform
Project Purpose	Replace manual library administration with a centralized digital platform for catalog, circulation, fines and reporting.
Objectives	Improve search speed, inventory accuracy, circulation efficiency and administrative visibility.
MVP Scope	RBAC, catalog, search, issue/return, fine calculation and basic reports.
Key Stakeholders	Students, faculty, librarians, library administrator, IT/DevOps and college management.
Milestones	M1 Domain understanding; M2 requirements and vision; M3 MVP and roadmap; M4 architecture/repository ready.
Success Criteria	Approved requirements, usable MVP definition, clear architecture and maintainable repository structure.
Key Risks	Scope creep, incomplete legacy data, changing library policies and insufficient user feedback.

14. Project Repository Initialization

The repository is prepared to support structured development and future Agile sprints.

- Create the project root folder and initialize Git.
- Add README.md describing purpose, setup and architecture.
- Add .gitignore for dependencies, environment files and build artifacts.
- Create initial branches: main, develop and feature/<name>.
- Commit the planning documents and initial project structure.

Suggested Initial Commit

```
git init

git add .

git commit -m "Initial commit: library platform planning and structure"
```

Repository Readiness

The repository should contain documentation, application modules, database definitions, tests and deployment configuration so development can proceed without mixing planning and implementation concerns.

15. Project Folder Structure

```
library-management-platform/
├── docs/
│   ├── BRD.md
│   ├── product-vision.md
│   ├── stakeholder-analysis.md
│   ├── personas.md
│   ├── FRS.md
│   ├── NFRS.md
│   ├── design-thinking-report.md
│   ├── roadmap.md
│   └── architecture.md
├── backend/
│   ├── src/
│   │   ├── modules/
│   │   │   ├── books/
│   │   │   ├── users/
│   │   │   ├── circulation/
│   │   │   ├── fines/
│   │   │   └── reporting/
│   │   └── common/
│   └── tests/
├── frontend/
│   ├── src/
│   │   ├── pages/
│   │   ├── components/
│   │   └── services/
│   └── public/
├── database/
├── docker/
├── kubernetes/
├── terraform/
├── monitoring/
└── README.md
```

15.1 Rationale

The structure separates documentation, frontend, backend, database and deployment concerns. Backend modules mirror the functional requirements, making independent testing and future Agile development easier.

16. Product Discovery Documentation

16.1 Discovery Methods

- Process study of book acquisition, cataloging, search, issue, return and overdue workflows.
- Stakeholder interviews or surveys with students, faculty and library staff.
- Observation of circulation queues and manual record-keeping.
- Problem framing using Design Thinking.

16.2 Key Findings

- Physical or fragmented searching increases the time required to locate resources.
- Paper checkout records create avoidable entry and reconciliation effort.
- Manual overdue tracking delays reminders and fine calculation.
- Administrators need centralized visibility into inventory and circulation activity.

16.3 Validated Problem Statement

Library members experience operational delays in locating, reserving, borrowing and returning resources because of inefficient manual tracking. At the same time, library staff lack centralized real-time visibility, resulting in record errors, inventory uncertainty and poor user experience.

16.4 Discovery Outcome

The findings directly inform the BRD, stakeholder analysis, personas, functional requirements, MVP scope and roadmap. The project therefore prioritizes search, circulation, inventory accuracy and overdue handling before advanced integrations and analytics.

Conclusion

The Digital Library Management Platform provides a complete Unit 1 product-planning foundation for a practical mini project. It begins with the business problem, identifies users and stakeholders, applies Design Thinking, defines functional and non-functional requirements, establishes an MVP, and prepares an architecture and repository for implementation.

The project is intentionally scoped so that the core library workflows can be implemented first and expanded later through Agile iterations. This follows the product-engineering approach in the reference material, where coding is one phase within a broader lifecycle involving requirements, architecture, UX, testing, deployment and continuous improvement.

Deliverables Completed

- Business Requirements Document
- Product Vision
- Stakeholder Analysis
- User Personas
- Functional Requirements Specification
- Non-Functional Requirements Specification
- Design Thinking Report
- IBM Design Thinking Report
- IBM Loop
- MVP Definition
- Product Roadmap
- System Architecture
- Project Charter
- Repository Initialization
- Project Folder Structure
- Product Discovery Documentation

— END OF MINI PROJECT —

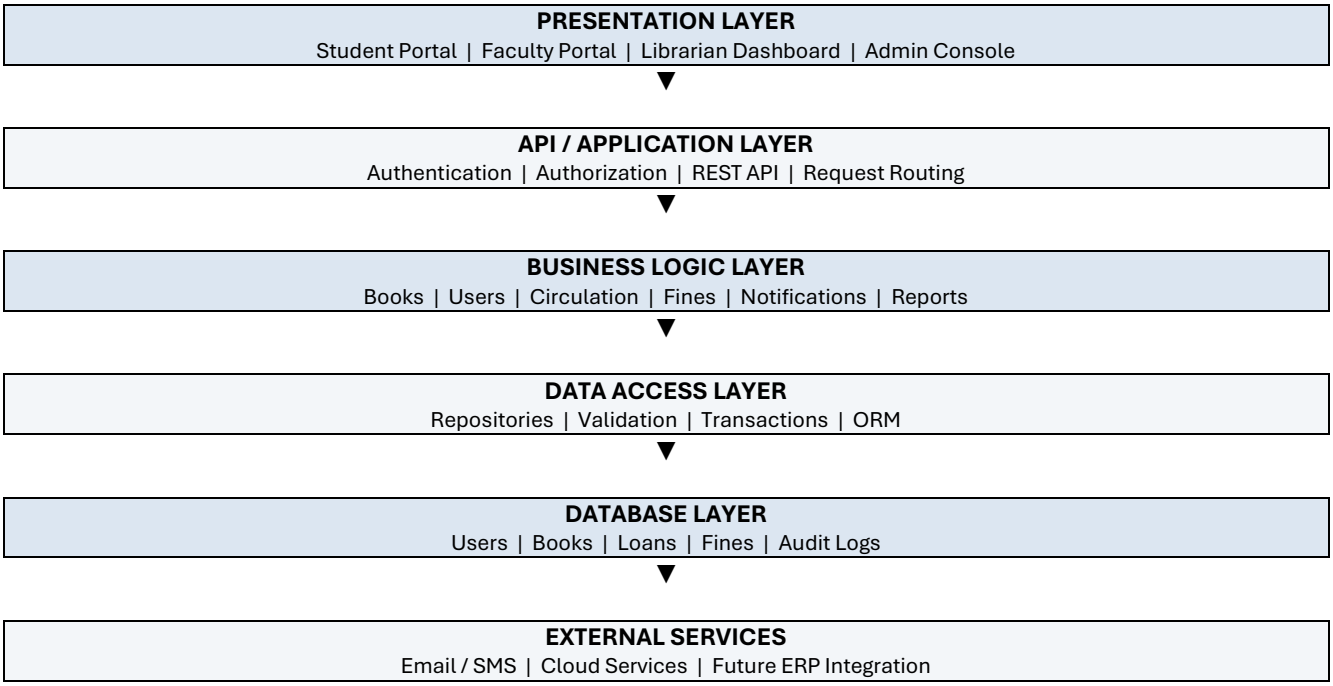
Visual Models, Flow Charts & Diagrams

The following visuals translate the requirements into easy-to-understand product, process, architecture and data views. They are included to make the mini project presentation clearer and more professional.

BOOK ISSUE / RETURN FLOW CHART

START	User requests a book
↓	
SEARCH	Search catalog
↓	
DECISION	Book available?
↓	
NO	Show unavailable / reservation option
↓	
YES	Verify member & loan eligibility
↓	
ISSUE	Create loan + reduce available copies
↓	
RETURN	On return, calculate overdue days
↓	
FINE	If overdue → calculate fine
↓	
UPDATE	Increase available copies + close loan
↓	
END	Transaction complete

SYSTEM ARCHITECTURE — LAYERED VIEW



DATABASE RELATIONSHIP DIAGRAM (ERD)

USERS	LOANS	BOOKS
user_id (PK) name role email	loan_id (PK) user_id (FK) book_id (FK) issue_date due_date return_date fine	book_id (PK) title author ISBN category total_copies available_copies
MEMBERSHIP		**CATEGORIES**
membership_id (PK) user_id (FK) status		category_id (PK) name
AUDIT_LOGS		**NOTIFICATIONS**

Key relationships: Users 1—M Loans; Books 1—M Loans; Books M—1 Categories; Users 1—1 Membership; Users 1—M Audit Logs; Users 1—M Notifications.

USE-CASE VIEW

STUDENT	LIBRARIAN	ADMIN
Search • View availability • Borrow • Renew	**Add books • Issue • Return • Manage fines**	**Users • Reports • Audit**
↓	↓	

Recommended UI Screens / Pictures

For the implemented prototype, the following screens should be designed as simple web pages: Login, Student Dashboard, Book Search, Book Details, Librarian Issue/Return, Fine Details, and Admin Reports. These screen ideas follow directly from the project's functional requirements and MVP scope.

Screen	Main Content	Primary User
Login	Email / PRN, password, role-based access	All
Student Dashboard	Search bar, borrowed books, due dates, fines	Student
Book Search	Filters, availability, author/category, details	Student / Faculty
Issue / Return	Member lookup, book lookup, transaction confirmation	Librarian
Admin Dashboard	Total books, issued, overdue, active members	Administrator
Reports	Inventory, circulation and overdue summaries	Administrator

Visual Summary

USER	WEB UI	REST API	BUSINESS SERVICES	DATABASE
↓	↓	↓	↓	

This visual summary shows the complete request flow from the user interface to the database, while the detailed architecture and process diagrams above explain each stage.